Traces of Orlicz-Sobolev functions under general growth restrictions and applications

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I came to know Mirek Krbec since about 1993. At that time I was a PHD student. I was invited to Prague by Professor Jana Stara and here our cooperation originated...

Joint papers:

1. Agnieszka Kałamajska, Miroslav Krbec, Gagliardo–Nirenberg inequalities in regular Orlicz spaces involving nonlinear expressions, J. Math. Anal. Appl. **362** (2010), 460–470.

2. Agnieszka Kałamajska, Miroslav Krbec, Traces of Orlicz-Sobolev functions under general growth restrictions, to appear in Mathematische Nachrichten, published online: 11 May 2012.

3. Agnieszka Kałamajska, Miroslav Krbec, Orlicz regularity theory for solutions to heat equation with the initial condition in Orlicz-Slobodetskii space, preprint 2012.

1 Besov Slobodetskii spaces. The classical definition and its role

This space appeared in papers by Aronszajn (1955), Slobodetski (1958), Gagliardo (1957).

Definition 1 *Let* $k > 0, k \notin \mathbf{N}$, $\lambda = k - [k]$, $\Omega \subseteq \mathbf{R}^n$ *is an open set. The function* $u : \Omega \to \mathbf{R}$ *belongs to Slobodetskii space* $W^{k,p}(\Omega)$ *if* $u \in W^{[k],p}(\Omega)$ *and*

$$\mathcal{N}_{\alpha,k,p} := \int_\Omega \int_\Omega \frac{|D^\alpha u(x) - D^\alpha u(y)|^p}{|x-y|^{n+p\lambda}} dx\, dy < \infty,$$

whenever $|\alpha| = [k]$.
It is a Banach space when equipped with the norm

$$\|u\|_{W^{k,p}(\Omega)} := \|u\|_{W^{[k],p}(\Omega)} + \sum_{\alpha,|\alpha|=[k]} \mathcal{N}^{\frac{1}{p}}_{\alpha,k,p}(u).$$

Example 1 *When* $k = 1 - \frac{1}{p}$ *we have* $[k] = 0$ *and* $u \in W^{k,p}(\Omega)$ *when* $u \in L^p(\Omega)$ *and*

$$\int_\Omega \int_\Omega \frac{|u(x) - u(y)|^p}{|x - y|^{p+n-1}} dx\, dy$$

$$= \int_\Omega \int_\Omega \left(\frac{|u(x) - u(y)|}{|x - y|} \right)^p \frac{1}{|x - y|^{n-1}} dx\, dy < \infty.$$

This space plays role in the trace embedding theorem.

- We are particularly interested in $W^{k,p}(\partial\Omega)$ where $0 < k < 1$, Ω is a Lipschitz boundary domain.
- In such case, when $\Omega \subseteq \mathbf{R}^n$: $u \in W^{1-\frac{1}{p},p}(\partial\Omega)$ when $u \in L^p(\partial\Omega)$ and

$$\int_{\partial\Omega}\int_{\partial\Omega}\left(\frac{|u(x)-u(y)|}{|x-y|}\right)^p \frac{1}{|x-y|^{n-2}}\sigma(dx)\,\sigma(dy) < \infty$$

with $n-1$-dimensional Hausdorff measure on $\partial\Omega$.

Theorem 1 *If $\Omega \subseteq \mathbf{R}^n$ is a bounded domain with Lipschitz boundary, $p > 1$, $k = 1 - \frac{1}{p}$ then we have*

i)(embedding theorem)

$$W^{1,p}(\Omega) \subseteq W^{k,p}(\partial\Omega), \qquad (1)$$
$$\|u|_{\partial\Omega}\|_{W^{k,p}(\partial\Omega)} \leq C\|u\|_{W^{1,p}(\Omega)};$$

ii)(extension theorem)

Any $u \in W^{k,p}(\partial\Omega)$ can be extended to $\tilde{u} \in W^{1,p}(\Omega)$ in such a way that

$$\tilde{u}|_{\partial\Omega} = u.$$

- (1) means that when $u_n \in C^\infty(\bar{\Omega}) \cap W^{1,p}(\Omega)$ and $u_n \to u$ in $W^{1,p}(\Omega)$ then $\{u_n\}$ is a Cauchy sequence in $W^{k,p}(\partial\Omega)$, so that

  $$u_k \to v \text{ in } W^{k,p}(\partial\Omega)$$

  for certain v. We define

  $$Tr(u) =: v \,(\text{“} = u|''_{\partial\Omega}).$$

  It does not depend on the choice of sequence $\{u_m\}$.

- ii) means $\tilde{u} \in W^{1,p}(\Omega)$ and $Tr(\tilde{u}) = u$.

2 Orlicz variant - setting of the problem and known results

2.1 Orlicz space

When $\Psi : [0, \infty) \to [0, \infty)$ (*Orlicz function*) is a nondecreasing convex continuous function such that $\Psi(0) = 0$ and $\lim_{t\to\infty} \Psi(t) = \infty$, the space

$$L^{\Psi}(\Omega) :=$$
$$\{f \in L^1_{\text{loc}}(\Omega) : \int_\Omega \Psi(s|f(x)|)\, dx < \infty \text{ for some } s > 0\}$$

is called the *Orlicz space*. It is a Banach space equipped with the Luxemburg norm:

$$\|f\|_{L^{\Psi}(\Omega)} := \inf\{\lambda > 0 : \int_\Omega \Psi\left(\frac{|f(x)|}{\lambda}\right) dx \leq 1\}.$$

When $\Psi(\lambda) = \lambda^p$ and $p > 1$, then $L^{\Psi}(\Omega) = L^p(\Omega)$.

- The same notation is used for vector-valued functions, $u : \Omega \to \mathbf{R^m}$, with the formal difference that instead of $|u(x)|$ we consider the Euclidean norm of $u(x)$.
- $\Psi : [0, \infty) \to [0, \infty)$ is a N-function if it is convex, $\lim_{\lambda \to 0} \frac{\Phi(\lambda)}{\lambda} = 0^+$ and $\lim_{\lambda \to \infty} \frac{\Phi(\lambda)}{\lambda} = \infty$
- we write: $\Psi \in \Delta_2$ if it satisfies the Δ_2-condition: $\Psi(2\lambda) \leq C\Psi(\lambda)$, for every $\lambda > 0$, with a constant C independent of λ.
- notation: $\Psi \in \Delta_2^c$ will mean that the Legendre conjugate of Ψ, that is, $\Psi^*(s) := \sup_{t>0}\{st - \Psi(t)\}$, satisfies the Δ_2-condition.

2.2 Orlicz-Sobolev space

We assume:

- $\Omega \subseteq \mathbf{R}^n$ be an open bounded domain,
- $\Psi : [0, \infty) \to [0, \infty)$ is Orlicz function.

Definition 2 *Let $k \in \mathbf{N}$. By Orlicz-Sobolev space $W^{k,\Psi}(\Omega)$ we will mean the linear set*

$$\{u \in L^1_{\mathrm{loc}}(\Omega) : D^\alpha u \in L^\Psi(\Omega) \text{ for every } \alpha : |\alpha| \leq k\},$$

equipped with the norm:

$$\|u\|_{W^{k,\Psi}(\Omega)} := \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^\Psi(\Omega)}.$$

2.3 Orlicz-Slobodetskii-type spaces

We assume:

- $\Psi, \Phi : [0, \infty) \to [0, \infty)$ are given two Orlicz functions.

Definition 3 *By $Y^{\Psi,\Phi}(\Omega)$ we denote the linear hull of all $u \in L^{\Psi}(\Omega)$, for which the modular*

$$I^{\Phi}(u, \Omega) := \int_{\Omega} \int_{\Omega} \Phi\left(\frac{|u(x) - u(y)|}{|x - y|}\right) \frac{1}{|x - y|^{n-1}} \, dxdy \tag{2}$$

is finite. We equip $Y^{\Psi,\Phi}(\Omega)$ with the norm

$$\|u\|_{Y^{\Psi,\Phi}(\Omega)} := \|u\|_{L^{\Psi}(\Omega)} + J^{\Phi}(u, \Omega), \tag{3}$$

where

$$J^{\Phi}(u, \Omega) := \inf\left\{\lambda > 0 : I^{\Phi}\left(\frac{u}{\lambda}, \Omega\right) \leq 1\right\} \tag{4}$$

is the Luxemburg seminorm generated by $I^{\Phi}(u, \Omega)$.

- By $Y_{\infty}^{\Psi,\Phi}(\Omega)$ we will mean the completion of set

$$\{u \in C^{\infty}(\bar{\Omega}) \cap Y^{\Psi,\Phi}(\Omega)\} \tag{5}$$

in the norm $\|u\|_{Y^{\Psi,\Phi}(\Omega)}$.

- One can define $Y^{\Psi,\Phi}(\partial\Omega)$ where $\Omega \subseteq \mathbf{R}^n$ is a Lipschitz boundary domain. Then instead of "dx'' one has to do with $\sigma(dx)$- the $n-1$-dimensional Hausdorff measure defined on $\partial\Omega$:

$$I^{\Phi}(u, \partial\Omega) :=$$
$$\int_{\partial\Omega}\int_{\partial\Omega} \Phi\left(\frac{|u(x)-u(y)|}{|x-y|}\right) \frac{1}{|x-y|^{n-1}}\,\sigma(dx)\sigma(dy)$$

Remark 1
If $\Omega \subseteq \mathbf{R}^n$ be an open domain with locally Lipschitz boundary, $\Psi(\lambda) = \Phi(\lambda) = |\lambda|^p$, $1 < p < \infty$. Then

$$\|u\|_{Y^{\Psi,\Phi}(\partial\Omega)} \sim$$

$$\|u\|_{L^p(\partial\Omega)} + \left(\int_{\partial\Omega} \int_{\partial\Omega} \frac{|u(x) - u(y)|^p}{|x - y|^{p+n-2}} \, dxdy \right)^{\frac{1}{p}},$$

which is the norm of u in the Slobodetskii space $W^{1-\frac{1}{p},p}(\partial\Omega)$.

2.4 Embedding end extension. Known results in Orlicz setting

A. Fougeres (1972), M-T. Lacroix (1974), G. Palmieri (1979):

Theorem 2 *When Φ is a N-function and both Φ and Φ^* satisfy the Δ_2-condition, then we have:*

i)(embedding)

$$W^{1,\Phi}(\Omega) \subseteq Y_{\infty}^{\Phi,\Phi}(\partial\Omega)$$

and the embedding is continuous;

ii) (extension) *The above embedding is a surjection.*

Question: What can be said if one relaxes assumptions:

- Φ is a N-function
- $\Phi \in \Delta_2, \Phi^* \in \Delta_2$.

The related works:

- A. Cianchi (2010): embeddings having the form

$$W^{k,A}(\Omega) \subseteq L^B(\partial\Omega)$$

where $k \in \mathbf{N}$, involving Orlicz functions A, B (optimal pairs A, B were found);

- A. Cianchi and L. Pick (2010): an approach dealing with Lorentz-Sobolev spaces $W^k L^{q,1}(\mathbf{R}^n)$ instead of $W^{k,q}(\Omega)$;

- A. Cianchi, R. Kerman and L. Pick (2008): an approach with embedding of the form

$$W^m X(\Omega) \subseteq Y(\partial\Omega)$$

where X, Y are rearrangement invariant spaces.

- It looks that since 1979 there is no progress when one asks about the embeddings into Orlicz extension of Slobodetskii space with the relaxed regularity assumption on the generating function.
- Such question seems to be important to understand better boundary value problems posed in Orlicz-Sobolev spaces.

3 Trace Theorem

3.1 The admissible spaces

Assumption A. We assume that $a, b : [0, \infty) \to [0, \infty)$ are strictly positive continuous functions such that

(a) $\int_0^1 a(s)/s\, ds < \infty$, $\int_1^\infty \frac{a(s)}{s}\, ds = +\infty$;

(b) $b(\cdot)$ is non-decreasing, $\lim_{s\to\infty} b(s) = +\infty$.

(c) there exist constants $c_1 > 0, s_0 \geq 0$ such that

$$\int_0^s \frac{a(t)}{t}\, dt \leq c_1 b(c_1 s) \text{ for all } s > s_0, \quad (6)$$

and in the case $s_0 > 0$ mapping $s \mapsto \frac{a(s)}{s}$ is bounded when $s \neq 0$ is close to 0.

We define

$$\Phi(t) := \int_0^t a(s)\, ds \quad \text{and} \quad \Psi(t) := \int_0^t b(s)\, ds, \quad (7)$$

where $t \geq 0$.

Remark 2 • We always have $\Phi(t) \leq \widetilde{C}\Psi(C_1 t)$, with universal constants, so that $\Phi \prec \Psi$.

- The following three statements a), b), c) are equivalent :
  a) $\Psi \in \Delta_2^c$
  b) Φ and Ψ are equivalent Orlicz functions,
  c) $\Phi \in \Delta_2^c$.

Example 2 *The pair:* $\Phi(\lambda) = \lambda \ln(2+\lambda)^{\alpha}$, $\Psi(\lambda) = \lambda \ln(2+\lambda)^{1+\alpha}$, $\alpha > 0$, *satisfies Assumption A.*

3.2 Main result

Theorem 3 (M. Krbec and A.K.)
Let Assumption A be satisfied and let Ω be a bounded domain with a locally Lipschitz boundary. Then trace operator $Tr : W^{1,\Psi}_{\infty}(\Omega) \mapsto Y^{\Psi,\Phi}_{\infty}(\partial\Omega)$ is well defined and there exists a constant D such that for every $u \in W^{1,\Psi}_{\infty}(\Omega)$ we have

$$\|Tru\|_{Y^{\Psi,\Phi}(\partial\Omega)} \leq D\|u\|_{W^{1,\Psi}(\Omega)}. \tag{8}$$

3.3 Proof of main result (the sketch)

We apply Hardy-type inequalities

Lemma 1 (Hardy-type inequality) *Let a, b and Φ, Ψ be as in Assumption A. Then*

i) *If (6) holds with $s_0 = 0$, then*

$$\int_a^b \Phi\left(\frac{1}{x-a}\int_a^x w(\xi)\,d\xi\right)\,dx \;\le\; C\int_a^b \Psi(w(x))\,dx,$$
$$\int_a^b \Phi\left(\frac{1}{b-x}\int_x^b w(\xi)\,d\xi\right)\,dx \;\le\; C\int_a^b \Psi(w(x))\,dx,$$

for every $w \ge 0, w \in L^1_{\mathrm{loc}}$, with a constant $C > 0$ independent of w.

ii) *If (6) holds with some $s_0 > 0$, then*

$$\int_a^b \Phi\left(\frac{1}{x-a}\int_a^x w(\xi)\,d\xi\right)\,dx \le C\left(\int_a^b w(x)\,dx + \int_a^b \Psi(w(x))\,dx\right),$$
$$\int_a^b \Phi\left(\frac{1}{b-x}\int_x^b w(\xi)\,d\xi\right)\,dx \le C\left(\int_a^b w(x)\,dx + \int_a^b \Psi(w(x))\,dx\right),$$

for every $w \ge 0, w \in L^1_{\mathrm{loc}}$, with a constant $C > 0$ independent of w.

Proof. It is based on generalizations of Hardy-Littlewood type inequatlities due to H. Kita (1997):

$$\int_{\mathbf{R}} \Phi\left(M(\widetilde{w}(\xi))\right)\, dx \;\leq\; C \int_{\mathbf{R}} \Psi(\widetilde{w}(x))\, dx,$$

where

$$M\widetilde{w}(x) := \sup_{x\in(c,d)} \frac{1}{d-c} \int_{(c,d)} \widetilde{w}(y)\, dy$$

is the Hardy-Littlewood maximal function of $\widetilde{w}$.

- when $\Phi \sim \Psi$ this is an extension of Hardy –Littlewood theorem dealing in the classical formulation with $\Phi = \Psi = \lambda^p$;

4 Extension operator

4.1 The classical approach

Let

- $Q^{'} = [0,1]^{n-1} \subseteq \mathbf{R}^{n-1}$ be the cube
- $u \in C_0^{\infty}(Q^{'})$.

Problem: Extend u to $\tilde{u} : Q = Q^{'} \times [0,1] \to \mathbf{R}$ in the way that

$$\|\tilde{u}\|_{W^{1,p}(Q)} \leq C\|u\|_{W^{1-\frac{1}{p},p}(Q')}.$$

The classical method: Take the regularization $\phi \in C_0^{\infty}(\mathbf{R}^{n-1})$, $\int \phi = 1$, supp$\phi \subseteq Q^{'}$ and define

$$\tilde{u}(x,\epsilon) := (u * \phi_{\epsilon})(x),\ x \in Q^{'}, \epsilon \in (0,1).$$

Clear: when $\epsilon \to 0$ we have: $\tilde{u}(x,\epsilon) \to u(x)$.

- the idea well works when $\Phi = \Psi = \lambda^p$
- also when $\Phi \sim \Psi$ (the Legendre conjugate of one of them satisfies the Δ_2-condition)

4.2 Our approach. Construction via heat kernel

We present our construction with the case when $\Omega = Q$, and assuming that $u \in C_0^\infty(Q')$. Let

$$\widetilde{E}(s,t) := \frac{1}{2^{n-1}(\pi t)^{(n-1)/2}} \exp(-s^2/4t),\ s \geq 0, t > 0,$$

$$E(x,t) := \widetilde{E}(|x|,t),\ x \in \mathbf{R}^{n-1},\ t > 0,$$

be the heat kernel.
Then we have:

1.

$$\int_{\mathbf{R}^{n-1}} E(x,t)\,dx = 1 \text{ for every } t > 0,$$

2.

$$\lim_{t \to 0} \int_{\mathbf{R}^{n-1}} \phi(x) E(x,t)\,dx = \phi(0)$$

for every $\phi \in C_0^1(\mathbf{R}^{n-1})$,

3.

$$E_t(x,t) = \Delta_x E(x,t).$$

We define

$$\tilde{u}(x,t) = (e^{t\Delta}u)(x) :=$$
$$\begin{cases} \int_{\mathbf{R}^{n-1}} u(y)E(x-y,t)\,dy = (E(\cdot,t) * u)(x) & t > 0, \\ u(x) & t = 0. \end{cases}$$

so that $\tilde{u}_t = \Delta\tilde{u}$

Lemma 2 *For any* $u \in C_0^\infty(Q')$ *we have* $Tr\widetilde{u} = u$.

4.3 Estimates of function $\tilde{u}$

Proposition 1 *For any $\lambda > 0$ and any convex function $R : [0, \infty) \to [0, \infty)$, we have*

$$\int_Q R\left(\frac{|\widetilde{u}(x', t)|}{\lambda}\right) dx'\, dt \le \int_{\mathbf{R^{n-1}}} R\left(\frac{|u(x')|}{\lambda}\right) dx'.$$

In particular, if R is a N-function and u is supported in Q', we have $\|\widetilde{u}\|_{L^R(Q)} \le \|u\|_{L^R(Q')}$.

Proposition 2 *Let (Φ, Ψ) satisfy Assumption A and*

$$\Phi(xy) \le D(1+G(x)\Phi(y)), \quad \int_0^1 G(|\log t|)\, dt < \infty,$$

with D independent of x, y.
Then

$$\int_{(x,t)\in Q} \Psi\left(\frac{|\widetilde{u}(x,t)|}{\lambda}\right) dxdt$$

$$\preceq 1 + \int_{Q'} \Phi\left(\frac{C|u(x)|}{\lambda}\right) dx, \text{ for every } \lambda > 0,$$

$$\|\widetilde{u}\|_{L^\Psi(Q'\times(0,1))} \preceq 1 + \|u\|_{L^\Phi(Q')}.$$

where constant $C > 0$ above is independent of u.

Example 3

$\Phi(\lambda) = \lambda \ln(2+\lambda)^{\alpha}$, $\Psi(\lambda) = \lambda \ln(2+\lambda)^{1+\alpha}$, $\alpha > 0$ are o.k.

4.4 Estimates of function $\nabla_x \widetilde{u}$

Proposition 3 *For any convex function R we have*

$$\int_{Q'} \int_{(0,1)} R\left(|\nabla_x \widetilde{u}|\right)\, dxdt \preceq$$
$$\int_{Q'} \int_{Q'} R\left(\frac{C_1|u(y)-u(x)|}{|x-y|}\right) \frac{1}{|x-y|^{n-3}}\, dxdy + \int_{Q'} R\left(C_2|u(x)|\right)\, dx,$$

In particular,

$$\int_{Q'} \int_{(0,1)} R\left(|\nabla_x \widetilde{u}|\right)\, dxdt \preceq$$
$$\int_{Q'} \int_{Q'} R\left(\frac{C_1|u(y)-u(x)|}{|x-y|}\right) \frac{1}{|x-y|^{n-2}}\, dxdy$$
$$+ \int_{Q'} R\left(C_2|u(x)|\right)\, dx.$$

Proposition 4 *Let* (Φ, Ψ) *satisfy Assumption A and* Φ *satisfies:*

$$\Phi(xy) \leq C(1 + G(x)\Phi(y)), \tag{9}$$

with C *independent of* x, y*, where* G *is non-increasing, locally bounded and such that*

$$\sup_{s<\frac{1}{2}} \frac{1}{s} \int_0^{s^2} \frac{G(|\log t|)}{|\log t|}\, dt < \infty.$$

Then for every $\lambda > 0$*:*

$$\int_{(0,1)} \int_{Q'} \Psi\left(\frac{|\nabla_x \widetilde{u}(x,t)|}{\lambda}\right) dxdt \preceq$$

$$1 + \int_{Q'} \int_{Q'} \frac{1}{|x-y|^{n-2}} \Phi\left(\frac{1}{\lambda}\frac{|u(x)-u(y)|}{|x-y|}\right) dxdy + \int_{Q'} \Phi\left(\frac{|u(x)|}{\lambda}\right) dx,$$

$$\|\nabla_x \widetilde{u}(x,t)\|_{L^\Psi(Q'\times(0,1))} \preceq \|u\|_{Y^{\Phi,\Phi}(Q')}.$$

Example 4 $\Phi(\lambda) = \lambda \ln(2+\lambda)^\alpha$, $\Psi(\lambda) = \lambda \ln(2+\lambda)^{1+\alpha}$, $\alpha > 0$ *are o.k.*

4.5 Estimates of function $\partial_t \tilde{u}$

A wishful thinking would expect the inequality of type

$$\int_{Q'} \int_{(0,1)} R\left(|\partial_t \tilde{u}(x,t)|\right)\, dxdt \preceq$$
$$\int_{Q'} \int_{Q'} R\left(\frac{|u(y)-u(x)|}{|x-y|}\right) \frac{1}{|x-y|^{n-2}}\, dxdy$$
$$+ \int_{Q'} R\left(|u(x)|\right)\, dx,$$

dealing with an arbitrary convex function R. It seems quite difficult to prove such an inequality by using our previous tools. As our first result in this direction we prove a weaker result, holding under more restrictive assumptions.

Assumption B:

1. $R(\lambda) = \lambda P(\lambda)$, where $P(ab) \prec 1 + P(a) + P(b)$,

2. P is nondecreasing,

3. Function $\frac{P(\lambda)}{\lambda}$ is nonincreasing for large arguments.

Example 5 *When* $P(\lambda) = (\log(2+\lambda))^{\alpha}$, *with any* $\alpha > 0$. *Then* P *satisfies* ***1, 2****, and* ***3****. Moreover,* $P(\lambda) \prec 1 + \lambda^{\beta}$, *with any* $\beta > 0$.

Proposition 5 *If R satisfies an assumption* **B**, *then*

$$\begin{aligned}
&\int_{Q'} \int_{(0,1)} R\left(|\partial_t \widetilde{u}(x,t)|\right)\, dxdt \\
&\preceq 1 + \int_{Q'} \int_{Q'} R\left(\frac{|u(y)-u(x)|}{|x-y|}\right) \frac{1}{|x-y|^{n-2}}\, dxdy \\
&\qquad\qquad + \int_{Q'} R\left(|u(x)|\right)\, dx.
\end{aligned}$$

Theorem 4 (M.Krbec, A.K) *Let* (Φ, Ψ) *satisfy Assumption* **A** *and* Φ *satisfies Assumption* **B**. *Then for every* $\lambda > 0$*:*

$$\int_{(0,1)} \int_{Q'} \Psi\left(\frac{|\partial_t \tilde{u}(x,t)|}{\lambda}\right) dxdt \preceq 1 + \int_{Q'} \int_{Q'} \frac{1}{|x-y|^{n-2}} \Psi\left(\frac{1}{\lambda}\frac{|u(x)-u(y)|}{|x-y|}\right) dxdy + \int_{Q'} \Psi\left(\frac{|u(x)|}{\lambda}\right) dx,$$

$$\int_{(0,1)} \int_{Q'} \Psi\left(\frac{|\partial_t \tilde{u}(x,t)|}{\lambda}\right) dxdt \preceq 1 + \int_{Q'} \int_{Q'} \frac{|\log|x-y||}{|x-y|^{n-2}} \Phi\left(\frac{1}{\lambda}\frac{|u(x)-u(y)|}{|x-y|}\right) dxdy + \int_{Q'} \Psi\left(\frac{|u(x)|}{\lambda}\right) dx,$$

$$\int_{(0,1)} \int_{Q'} \Psi\left(\frac{|\partial_t \tilde{u}(x,t)|}{\lambda}\right) dxdt \preceq 1 + \int_{Q'} \int_{Q'} \frac{1}{|x-y|^{n-2+\varepsilon}} \Phi\left(\frac{1}{\lambda}\frac{|u(x)-u(y)|}{|x-y|}\right) dxdy + \int_{Q'} \Psi\left(\frac{|u(x)|}{\lambda}\right) dx.$$

Number $\varepsilon > 0$ *in the last inequality can be taken arbitrarily.*

THANK YOU...